Factor Investing in Emerging Markets

From Multi-Factor stock selection to Market-Neutral Alpha

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EM equities offer a large, under-researched alpha opportunity

The Opportunity

- $7.5T market cap across 24 countries — structural inefficiencies persist
- Lower analyst coverage → slower price discovery → factor premia
- Diversification: low correlation (ρ = 0.22) with US factor returns

The Challenge

- Higher transaction costs (50–100 bps) erode naive factor strategies
- Currency risk and political instability add volatility layers
- Most research uses US/DM data — EM-specific validation is scarce

Our Result

- Hedged Sharpe 0.83 (net 0.67) with 4.5% annualized alpha in true OOS
- Max drawdown compressed from -27% to -5% via dynamic beta hedge
- Profitable to 175bp TC; US diversifier (ρ = 0.22) clears hurdle by 0.64

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Forget the 'Sell America' Trade. Here Comes 'Hedge America.'

U.S. stocks have rallied, but the dollar's slide and slowing foreign purchases of Treasurys are warning signs

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Research Question

Can a systematic, multi-factor long-only strategy generate persistent risk-adjusted alpha in EM equities after transaction costs and with practical constraints?

- Which factors work in EM? How should they be combined?
- Can dynamic beta hedging enhance risk-adjusted returns?
- Does the strategy survive realistic transaction costs?
- Does EM alpha diversify a US factor portfolio?

98,742 stock-month observations across 953 stocks and 11 industries

98,742	953	255	11
Observations	Stocks (Top 500 by mkt cap)	Months (2004–2025)	GICS Industries

Six Factor Signals

Size Log market cap — larger EM stocks outperform

Value (P/B) High P/B (growth) wins — growth premium in EM

Quality (ROE) Return on equity — lagged 1 quarter

Momentum (11m) Cumulative return from t-12 to t-2 — skipping most recent month

Volatility 24-month return std dev — low-vol stocks outperform

Dividend Yield Low yield outperforms — growth & reinvestment premium in EM

Data Cleaning Pipeline

Winsorization Cap monthly outliers at 1st/99th percentiles in returns and fundamentals

Country neutralization Remove country-level effects

Z-scoring Standardize factors within each industry-month for comparability

Median imputation Fill missing factor values with the industry-month median

Temporal alignment Use factors at t to predict returns at t+1 — no lookahead bias

Data: Refinitiv Datastream, Bloomberg | Benchmark: MSCI EM Index | Ann. Return 6.5% | Ann. Vol 20.3% | Sharpe 0.34 | Max DD 62.7%

Strict 4-period temporal separation prevents any lookahead bias

Warmup

2004–2008
(60 months)

Cross-sectional factor construction & rank signals based on rolling 60m Rank IC

→

Factor Selection

2009–2013
(60 months)

Rank IC analysis → select factors & composition method → 11 multi-factor industry portfolios

→

PC Selection

2014–2018
(60 months)

Portfolio construction method selection & TC optimization → 1 long-only portfolio

→

True Holdout

2019–2025
(63 months)

Final out-of-sample evaluation & rolling 60m beta hedge with iShare EEM → market-neutral portfolio & performance attribution

Key design: No information from later periods ever leaks into earlier decisions

01

Factor Selection

Which factors predict EM cross-sectional returns?

Size and value deliver the strongest predictive signals across EM equities

Average Rank IC by Factor

Pooled Across 11 Industries, 2004–2025

Factor	Avg IC	|IC|	IC/SD	t-stat
Size (log_mktcap)	+0.032	0.032	0.184	**2.84 ★**
Value (P/B)	+0.028	0.028	0.145	**2.26 ★**
Momentum (11m)	+0.017	0.017	0.081	1.24
Quality (ROE)	+0.012	0.012	0.067	1.07
Dividend Yield	−0.017	0.017	−0.085	−1.27
Volatility	−0.013	0.013	−0.057	−0.84

★ Statistically significant at 5% level (t > 1.96)

Key Findings

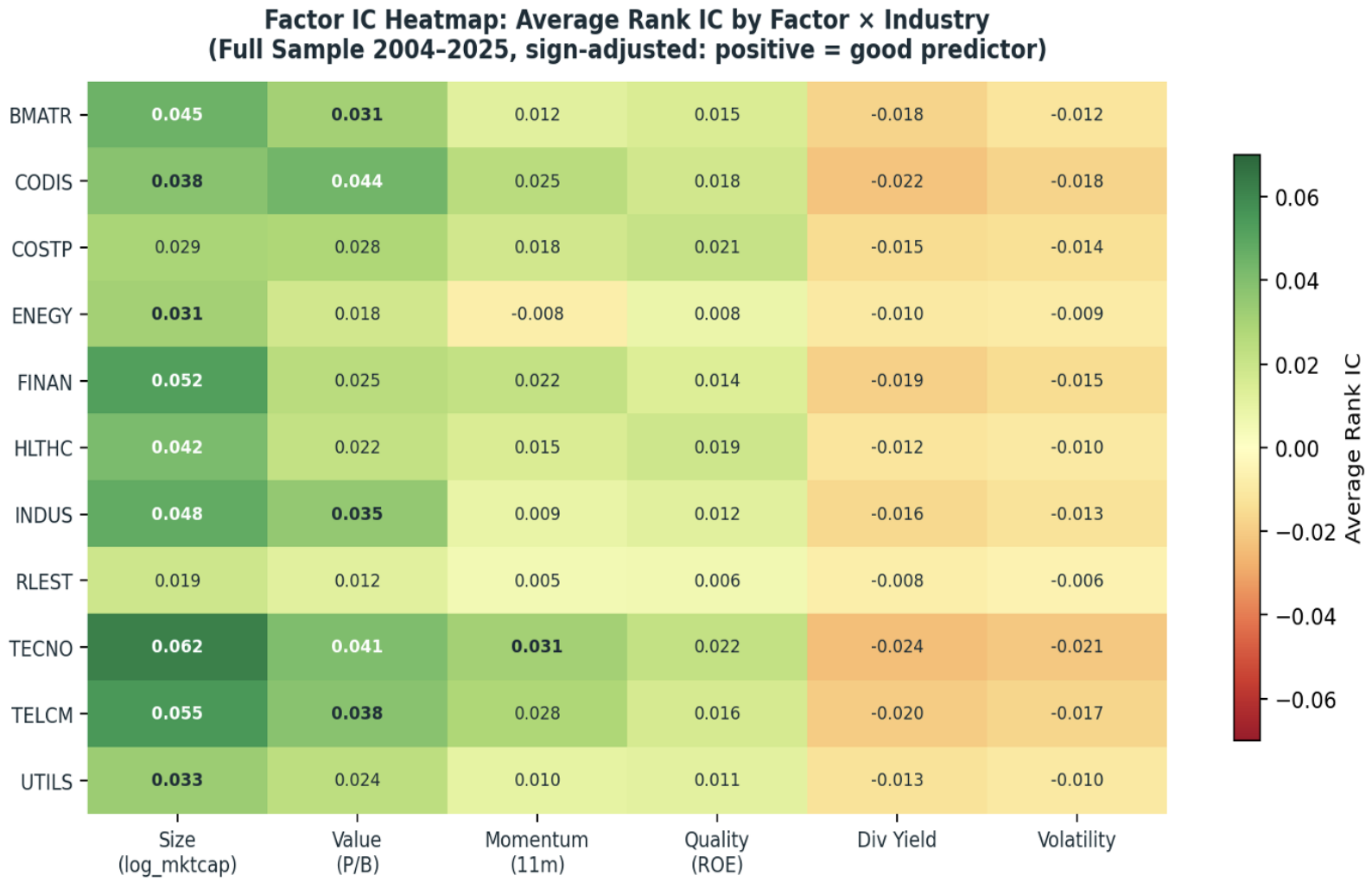
Size leads predictability Strongest IC (0.032, t=2.84). Reflects liquidity premium of large EM firms.

Value premium reverses in EM High P/B (growth) outperforms — opposite of developed markets. Consistent with MSCI EM Value underperformance.

Low-vol and low-yield win Negative ICs: low-volatility and low-dividend-yield stocks outperform — growth reinvestment premium in EM.

Data: Refinitiv Datastream, Bloomberg | Spearman rank IC pooled across 11 GICS industries | Full sample: 2004–2025 (255 months)

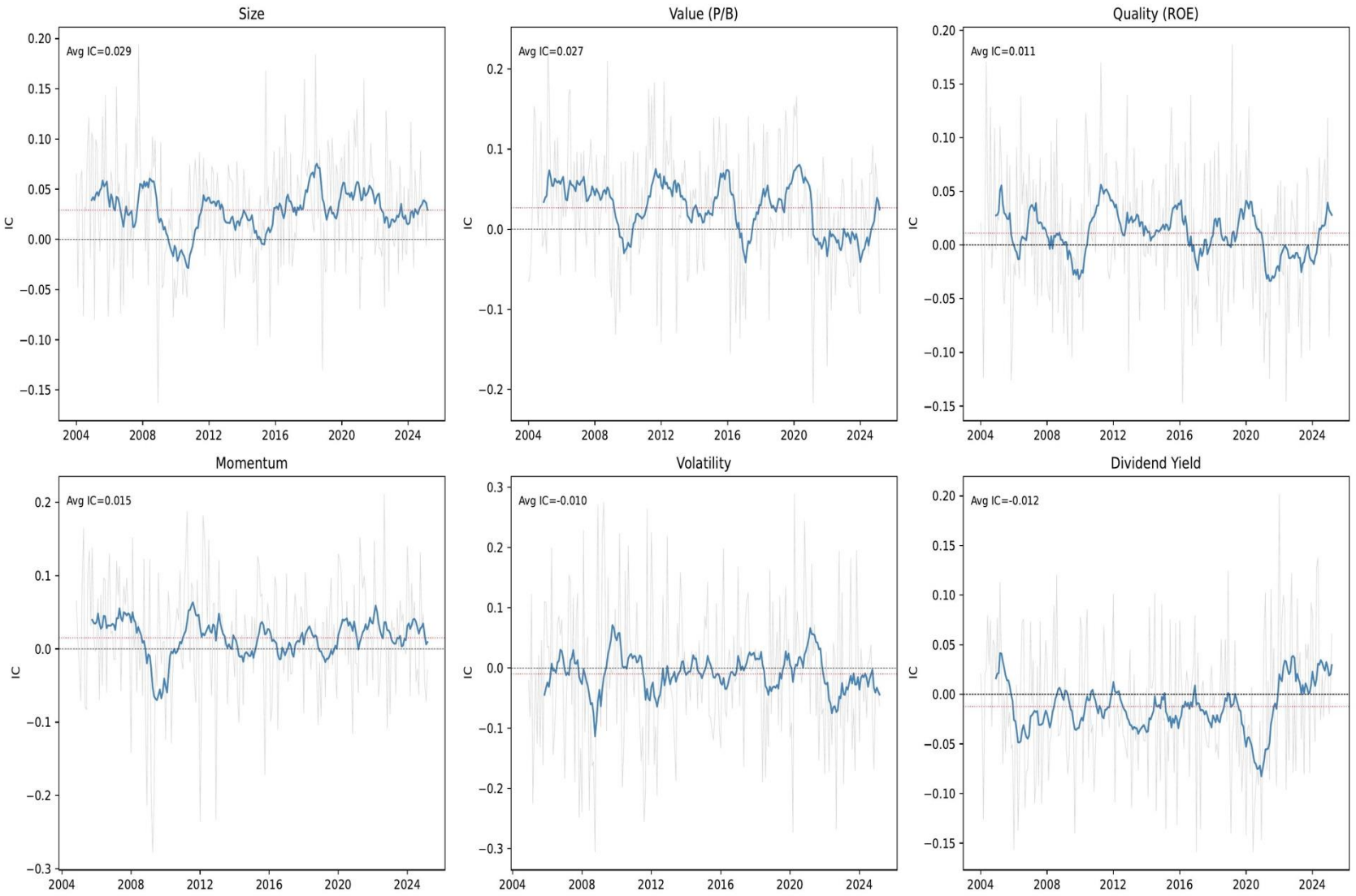
Factor predictability is consistent across all 11 GICS industries



Key Findings

- Size and Value show positive IC across all 11 GICS industries
- Technology exhibits the strongest factor signals
- Defensive factors (Volatility, Div Yield) show weak negative IC as expected
- **Cross-sectional breadth validates factor universality in EM**

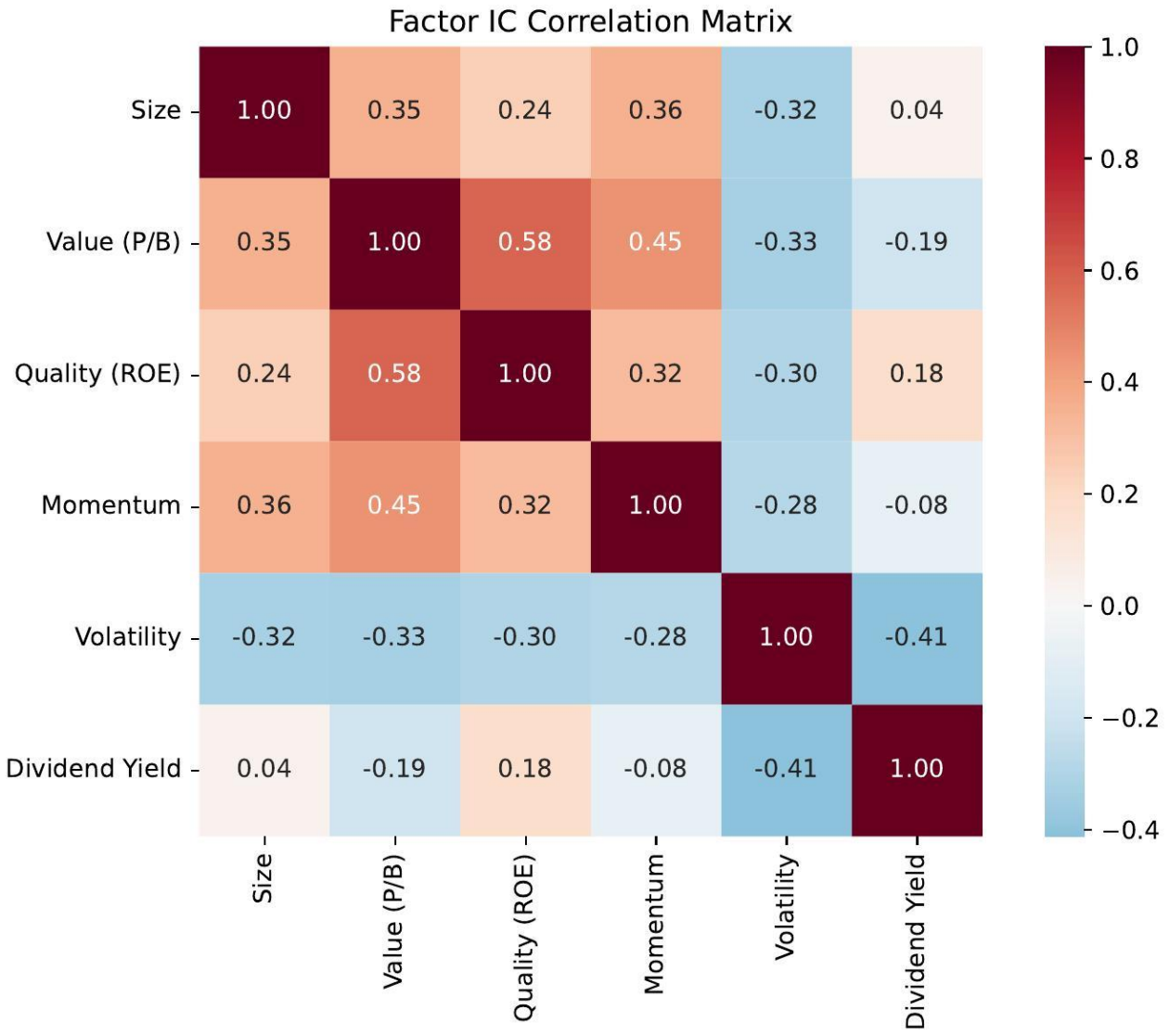
Rolling IC confirms factor signals persist through EM market cycles



Key Findings

- Size and Value IC remain positive across full 2004–2025 sample
- Momentum IC is episodic — high during trends, reverses in crises
- Volatility and Div Yield IC are noisy but mean-reverting near zero
- **No single-period dependence — signals persist through market cycles**

Low inter-factor IC correlation supports equal-weight diversification



Key Findings

- Most factor pairs show low IC correlation ($|\rho| < 0.4$)
- Value and Quality most correlated ($\rho \approx 0.6$)
- Size nearly uncorrelated with Dividend Yield factors
- **Low correlation → multi-factor composite has diversification benefit**

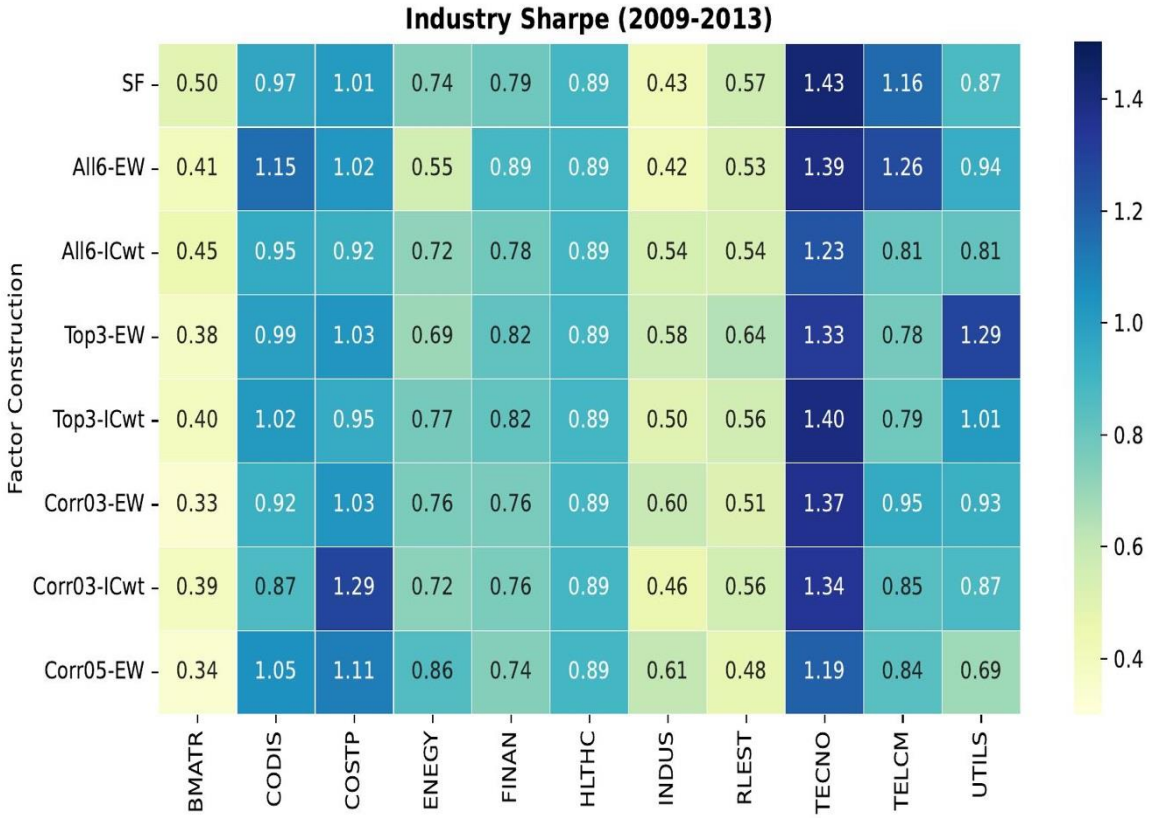
02

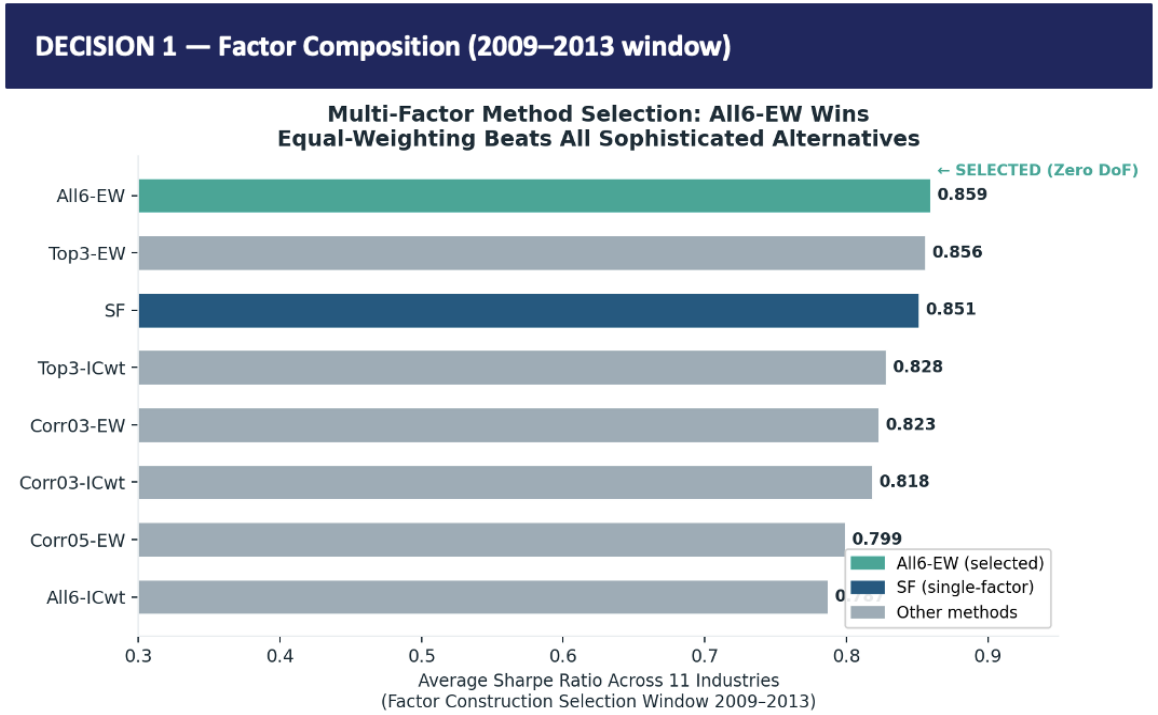
Multi-Factor & Portfolio Construction

How should we combine factors into a single signal?
Which Portfolio Construction method optimizes risk-adjusted returns after costs?

Multi-factor Industry Portfolio: All6-EW ranks #1 across 8 composition methods

Factor Selection Period 2009–2013 | 8 methods × 11 GICS industries





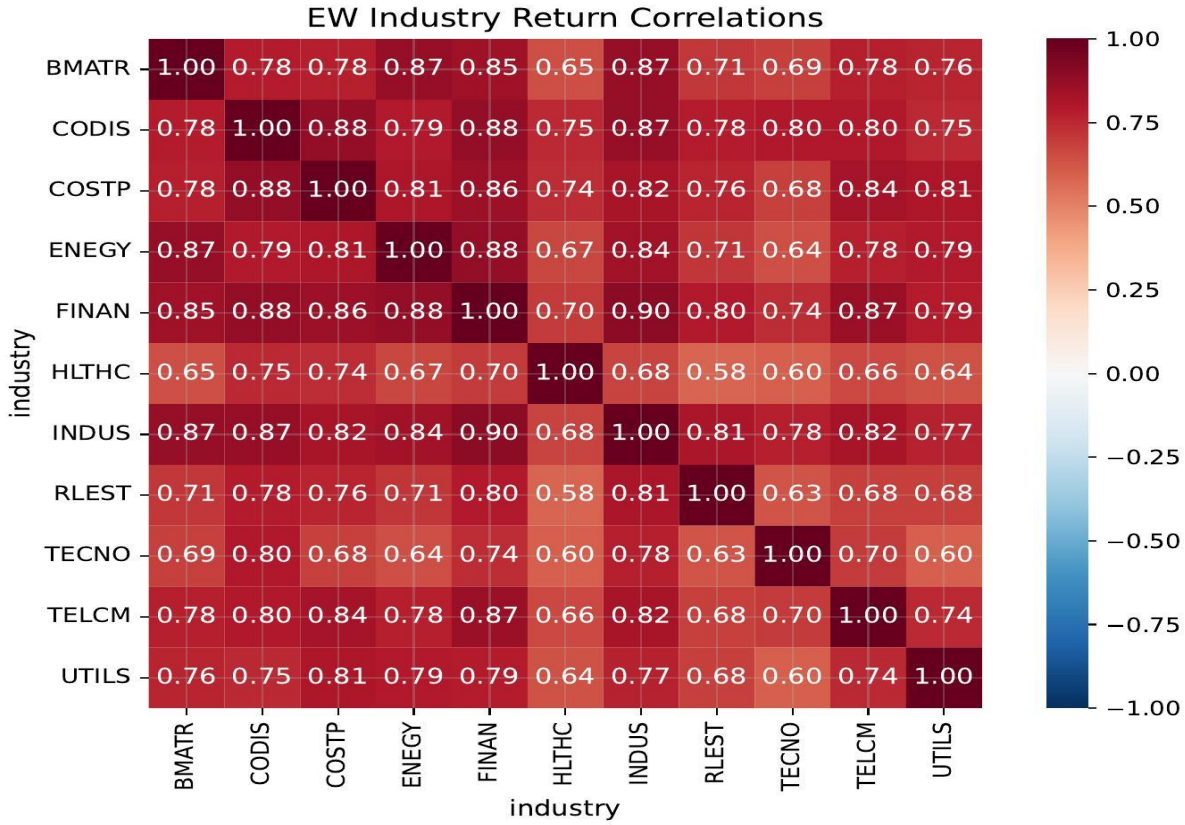
Practitioner Implication

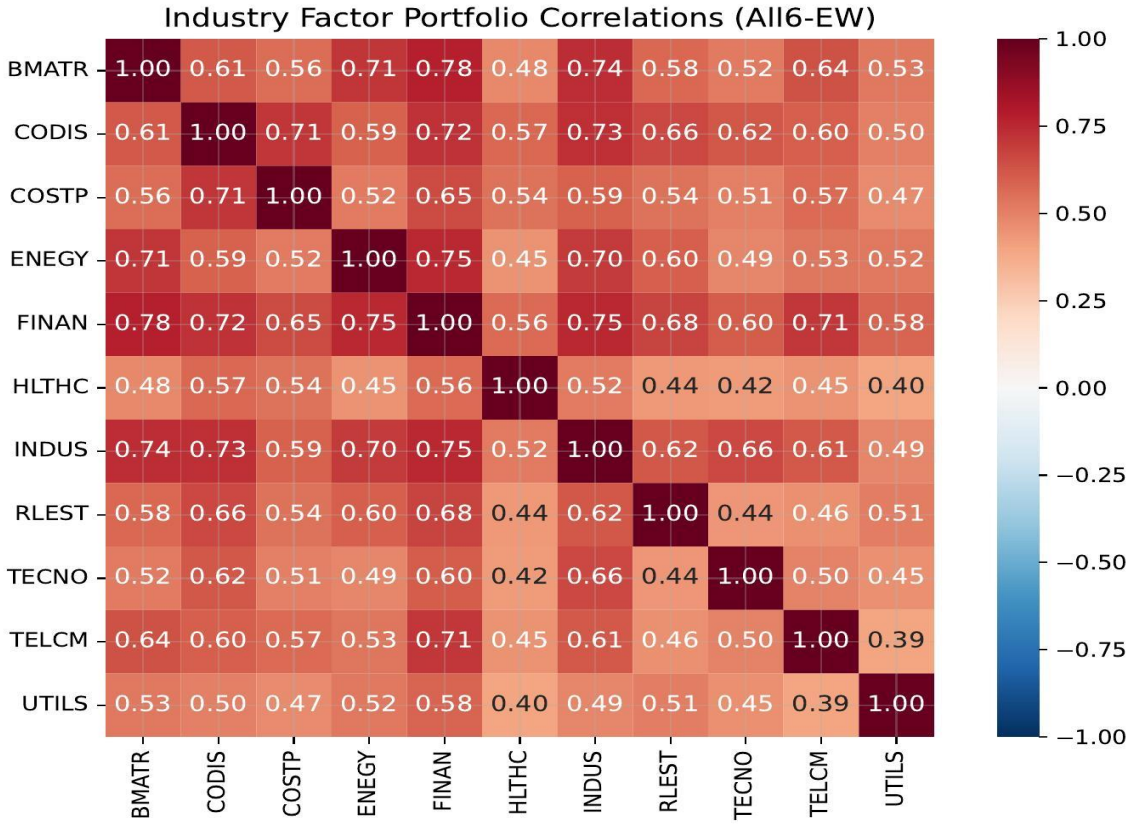
Simplicity wins in EM — complex models overfit limited data

- **Left: Sharpe ratios by method × industry. Right: rank within each industry. All6-EW averages ranked consistently top across sectors.**

Source: 60 months, 8 composition methods, zero degrees of freedom for All6-EW

Factor return correlations confirm diversification benefits of equal weighting





- Return correlations are modest across all factor pairs
- No single factor dominates — combining all 6 reduces strategy-specific risk
- Equal weighting: robust to misspecification of factor return forecasts

Source: Pairwise Pearson correlation of industry factor portfolio returns | Validation Sample 2014–2018

Turnover-penalized MVO delivers the highest net Sharpe across 10 methods

Objective Function (TO_MVO)

$$\max \; w'\mu - \lambda/2 \cdot w'\Sigma w - \kappa \cdot \|w - w_{t-1}\|_1$$

return − risk penalty − turnover penalty

Parameters & Constraints

λ Risk aversion — set at 2.5

κ Turnover penalty (γ) — set at 0.001

Long-only, fully invested, position bounds [2%, 20%]

Degrees of freedom: 3 (λ, κ, rebalance frequency)

Key Performance (Development 2014–2018)

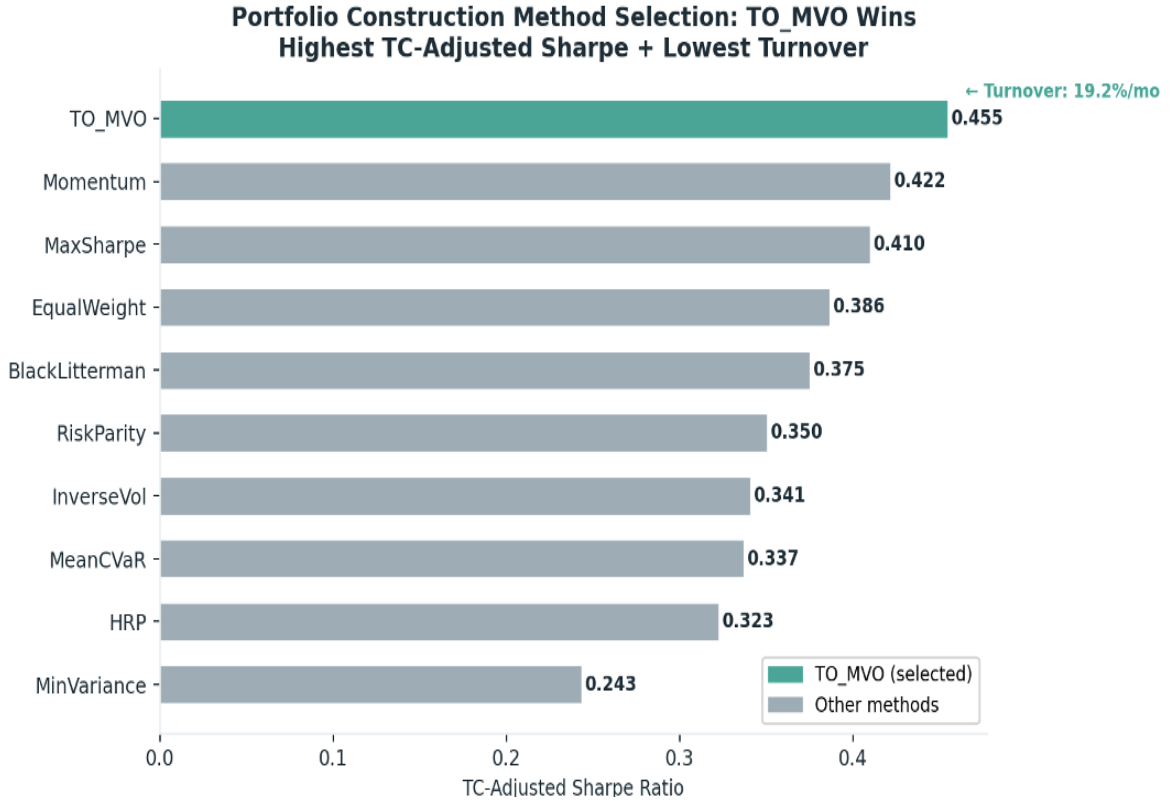
0.455
TC-Adj Sharpe (#1 of 10)

19.2%
Monthly Turnover

60%
TC Drag Reduction vs MVO

316 bps
Break-even Cost | Stability: 0.848

DECISION 2 — Portfolio Construction (2014–2018 window)



03

OOS Performance Evaluation

True out-of-sample test: 2019–2025 (never used in model selection)

True holdout confirms long-only strategy beats EEM by +490 bps annually

Long-Only Performance | Holdout: Jan 2019 – Sep 2024 | TC assumption: 45 bp round trip

Strategy	Ann. Return (Gross)	Ann. Return (Net, 45bp)	Ann. Vol	Sharpe (Gross)	Sharpe (Net)	Max DD	% Pos Mo
All6-EW + TO_MVO	**11.5%**	**10.5%**	**19.0%**	**0.607**	**0.552**	**−27.4%**	**60.9%**
EEM total return	6.6%	6.6%	18.5%	0.356	0.356	−36.7%	56.5%
MSCI EM (price index)	5.0%	5.0%	18.3%	0.274	0.274	−38.4%	56.5%

+55%
Sharpe improvement vs. EEM
(0.552 vs. 0.356)

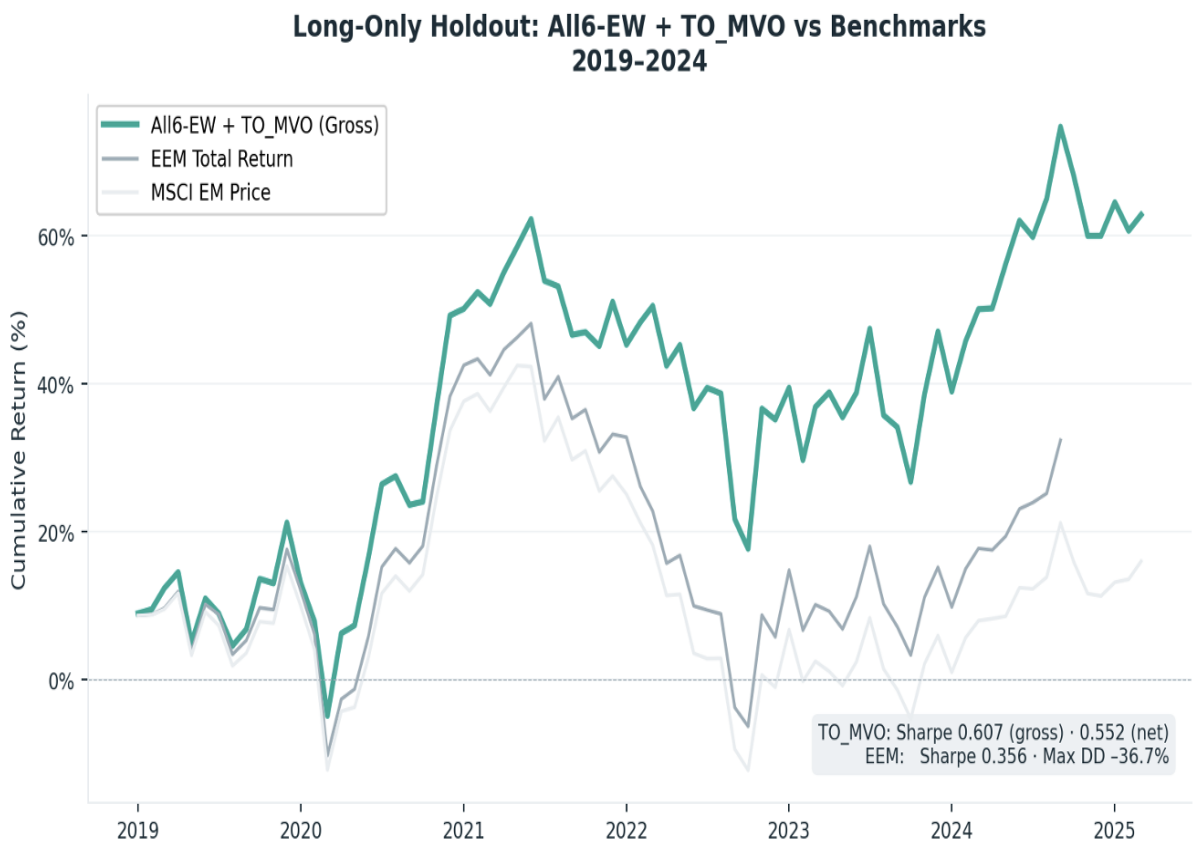
+490 bps
Excess annual return vs. EEM
after transaction costs

19.4%/mo
Holdout turnover vs.
19.2% in dev — identical

5 of 6
Calendar years beating
EEM benchmark

Source: True out-of-sample Jan 2019 – Sep 2024 | No parameters re-tuned in holdout

Strategy outperforms EEM benchmark through COVID crash and recovery



COVID Crash (Feb–Apr 2020)
Deep EM market selloff. Strategy outperforms — value and quality tilt provides natural downside cushion.

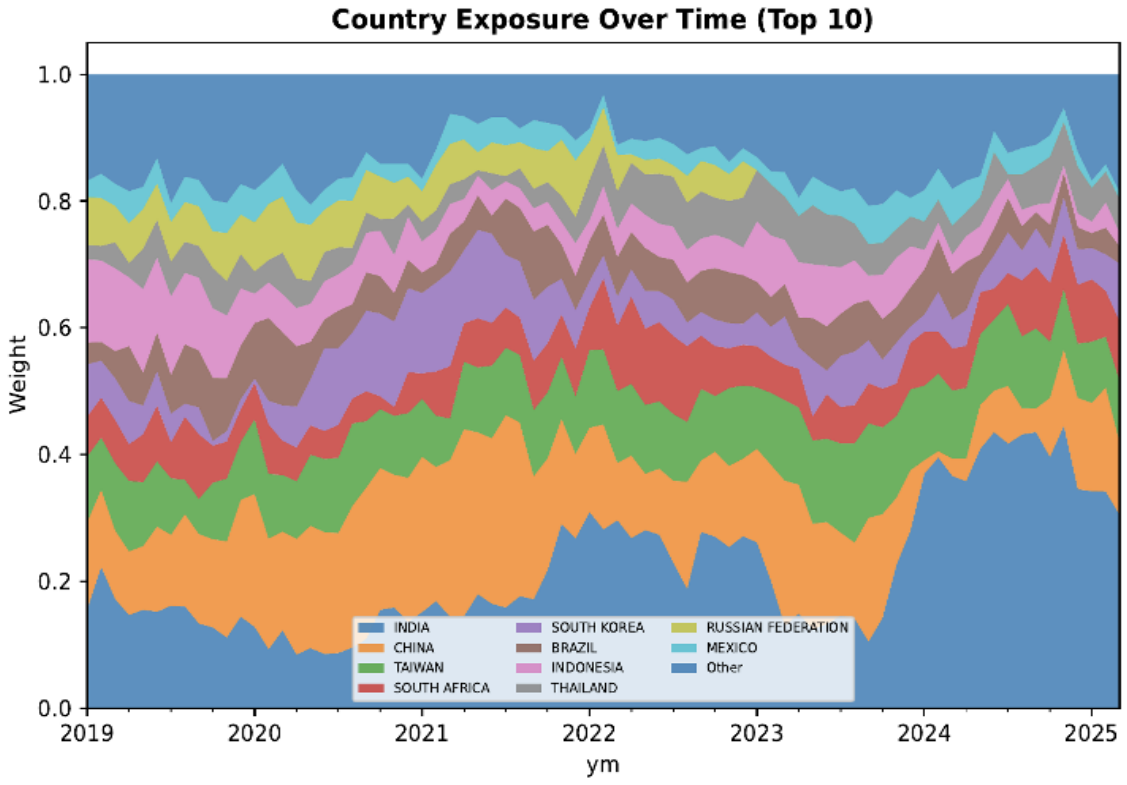
Post-COVID Rally (2020–2021)
EM upside captured. Momentum factor activates. Both gross and net returns strong.

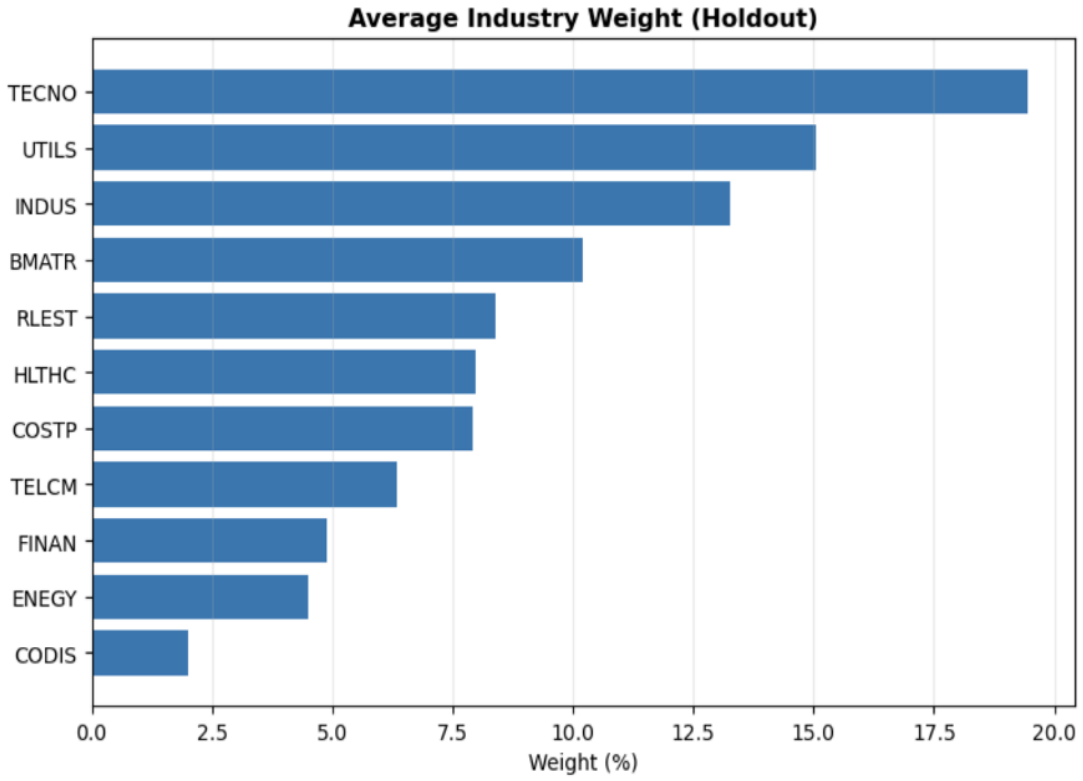
Inflation Shock (2021–2022)
Fed tightening, USD strengthening, EM under pressure. Strategy holds up vs. benchmark.

EM Recovery (2023–2025)
Sustained outperformance. Factor signals remain predictive across new macro regime.

Source: Cumulative net returns, All6-EW + TO_MVO vs. EEM | Holdout period 2019–2025

Technology leads industry allocation; country exposure rotates dynamically

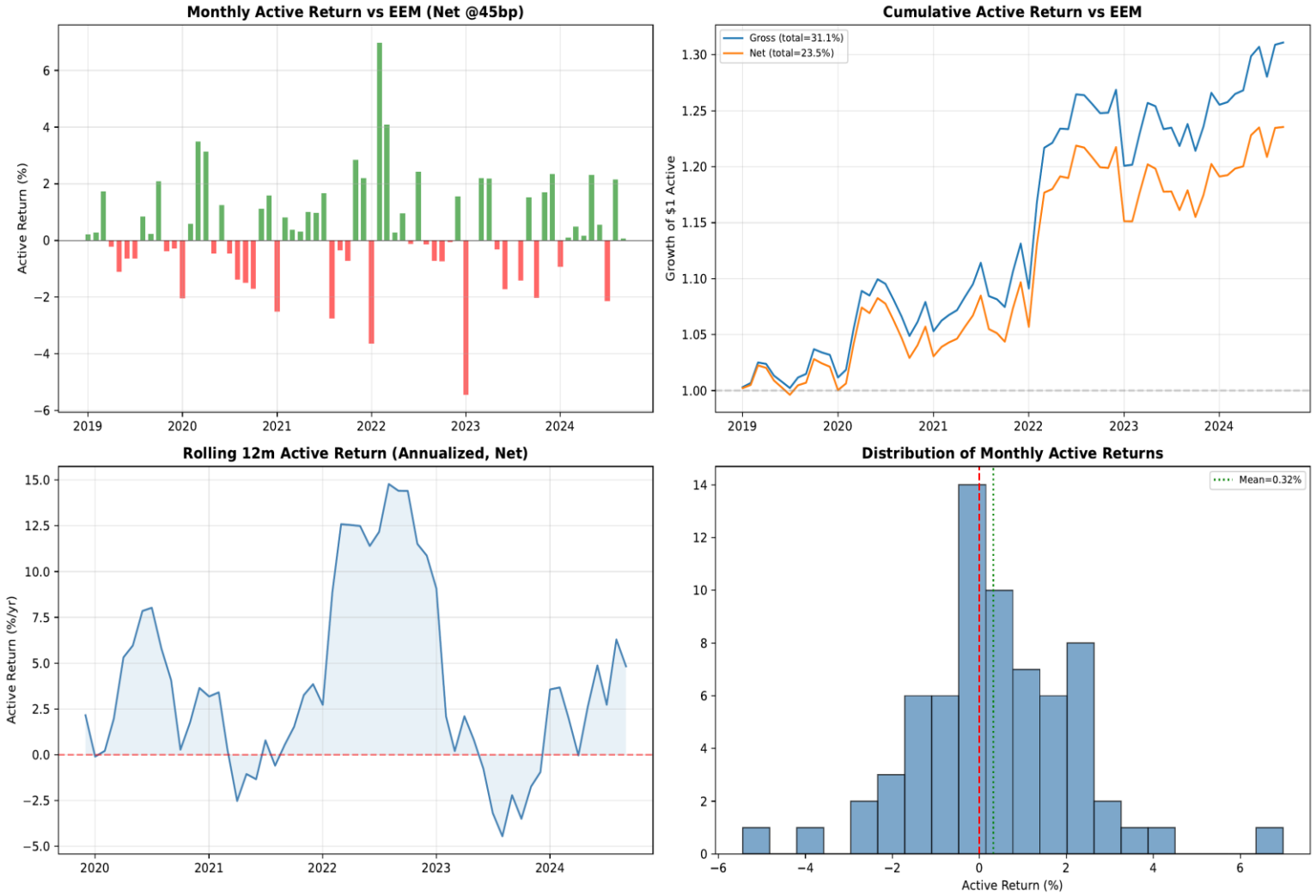




- **Country: allocation shifts dynamically over 2019–2024, reflecting time-varying factor signals rather than static country bets**
- **Industry: Technology dominates (~20%), followed by Utilities and Industrials — Financials and Consumer Discretionary significantly underweight vs benchmark**

Source: Average portfolio weight vs benchmark weight by country and industry, Holdout 2019–2024

Jensen’s alpha validation, motivating for dynamic beta hedge



Key Findings

- Rolling Jensen’s alpha remains consistently positive across the full holdout period, confirming persistent skill beyond market beta
- Alpha distribution is right-skewed — upside months contribute more than downside months detract, a favorable asymmetry
- Positive alpha net of market exposure validates beta-hedged construction — returns are not merely a reward for bearing EM market risk

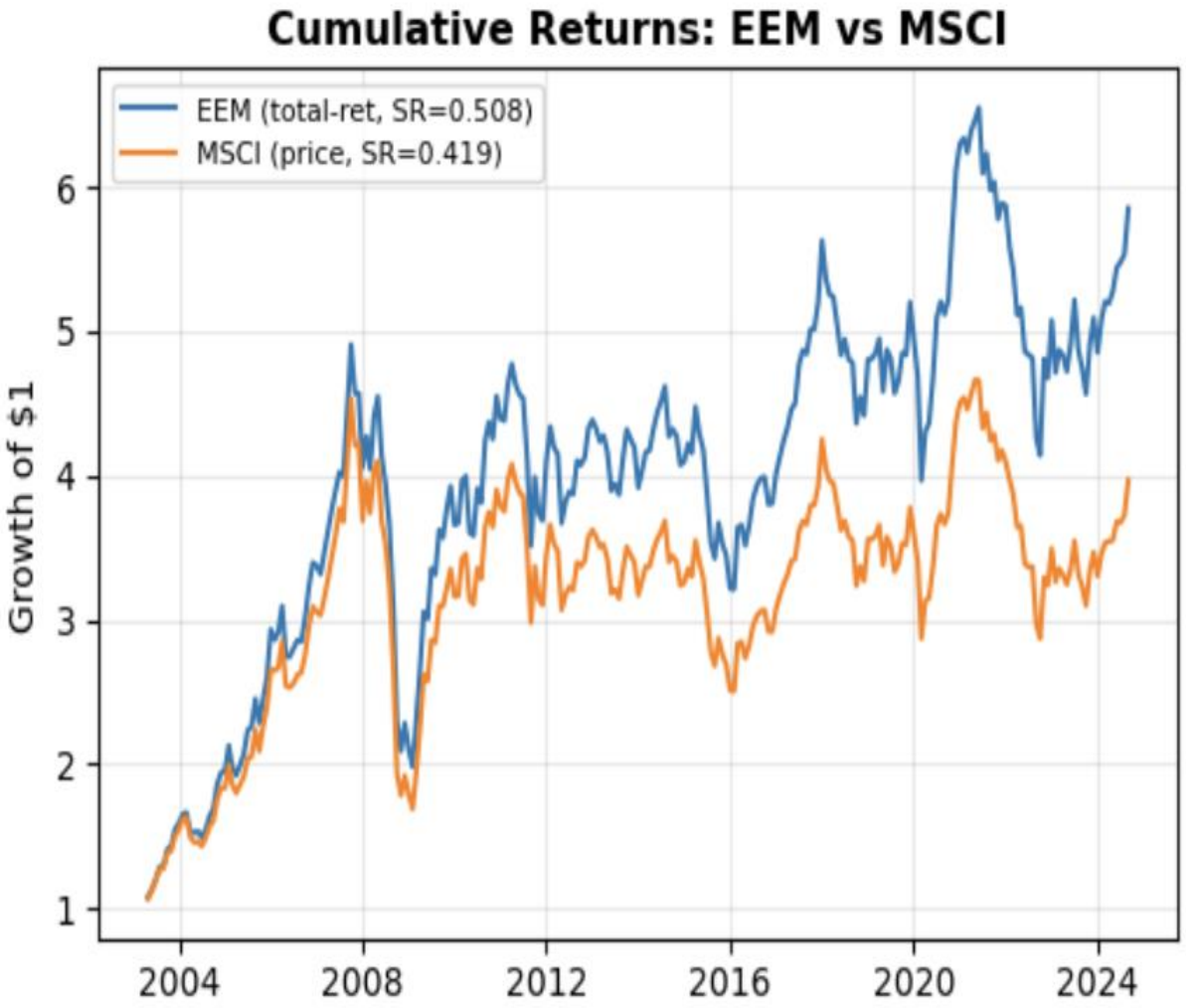
Source: Jensen’s alpha from CAPM regression vs. EEM | Holdout period 2019–2025

04

Dynamic Beta Hedging

Extracting pure alpha by neutralizing market exposure

We select short EEM as the hedge instrument over EM futures for cost and FX reasons



Short iShare EEM (ETF)

Pros

- Highly liquid, easy to short — deep borrow market
- Natural FX hedge —shorting offsets FX risk in the long-only leg
- Potential short rebate income on borrowed shares

Cons

- Foregone dividend yield on EEM (~2% p.a.)

Short MSCI EM Futures

Pros

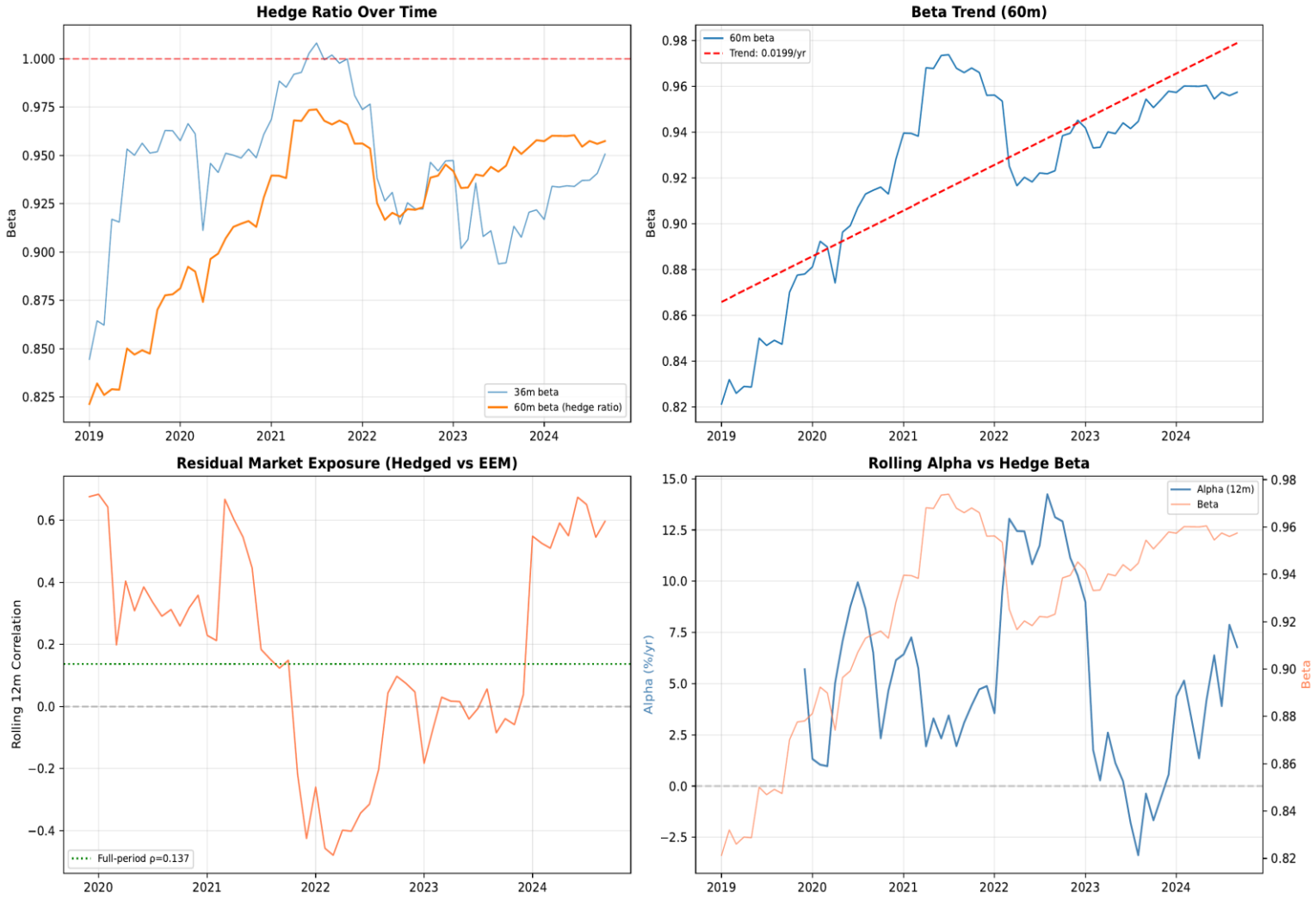
- Capital efficient — margin-based, frees up cash
- No dividend drag on the short leg

Cons

- EM forward discount erodes ~1.7% p.a. in USD return (carry cost)
- Roll cost and basis risk at contract expiry
- Does not offset FX exposure in the long-only portfolio

Source: Hedge instrument comparison | Dynamic beta hedge vs. EEM, holdout period 2019–2025

Hedge ratio adjusts through COVID, recovery, and 2022 drawdown

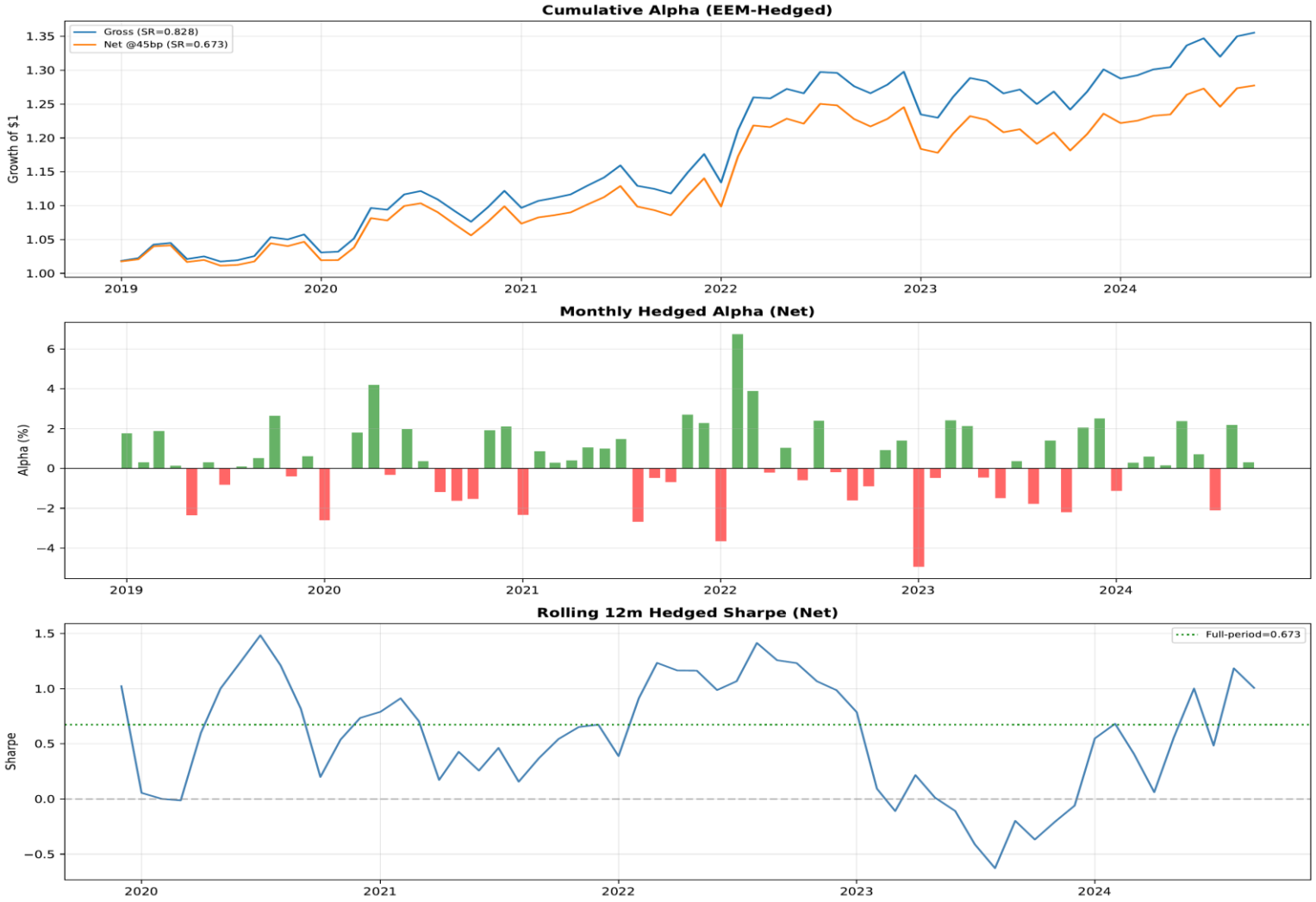


Hedging Mechanism

- Rolling 60-month OLS beta vs EEM total return
- Monthly short EEM sized to beta — hedge ratio adjusts dynamically
- Captures pure stock-selection alpha by removing market beta noise
- Beta is not constant — static hedging would leave significant market exposure unhedged during volatile periods

Source: Dynamic hedge ratio and cumulative P&L from short EEM position

Cumulative alpha reaches 4.48% annualized net of transaction costs



Key Performance (Out of Sample)

- **Net Sharpe Ratio: 0.673** — Improved 22% from long-only (0.552) after 45 bps TC
- **Net Alpha: 4.48% annualized** — Hedged portfolio return
- **Residual Volatility: 6.66%** — Down 65% from 18.9% long-only
- **Max Drawdown: −5.2%** — vs −27.4% long-only

Source: Holdout 2019–2024 | EEM hedge, TC = 45 bps | Alpha = portfolio return − β × EEM return

Strategy loads on CMA (β=0.43) despite no explicit investment factor input

DEPENDENT VARIABLE: TO_MVO EEM-Hedged Return – OOS (2019–2024)

	CAPM	FF3	FF5	FF5+Mom	FF3+Mom	CAPM+Mom
const	0.0042*	0.0038	0.0040*	0.0020	0.0020	0.0026
	(0.0023)	(0.0026)	(0.0023)	(0.0024)	(0.0027)	(0.0023)
Mkt_RF	0.0775	0.0812	0.1174**	0.1637***	0.1182**	0.1135**
	(0.0487)	(0.0579)	(0.0553)	(0.0561)	(0.0587)	(0.0507)
SMB		0.0515	0.1333	0.0986	0.0246	
		(0.1846)	(0.1869)	(0.1795)	(0.1803)	
HML		0.0428	-0.1628	-0.1611	0.0665	
		(0.1232)	(0.1403)	(0.1324)	(0.1252)	
RMW			0.1935	0.1724		
			(0.2059)	(0.1932)		
CMA			0.3871**	0.4304**		
			(0.1728)	(0.1730)		
Mom				0.1692*	0.1429	0.1350
				(0.0996)	(0.0974)	(0.0983)
R-squared	0.0408	0.0475	0.1229	0.1709	0.0826	0.0731
R-squared Adj.	0.0265	0.0035	0.0532	0.0907	0.0253	0.0450
N	69	69	69	69	69	69
R-squared	0.041	0.047	0.123	0.171	0.083	0.073
R-squared Adj.	0.026	0.004	0.053	0.091	0.025	0.045

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

Key Findings

- **α (Hedged):** marginally significant under CAPM and FF5 model with positive point estimates across all models
- **Factor Exposure:** no significant loading on SMB, HML, RMW, or Mom — only Market and CMA
- **Market β: 1.07 → 0.16** — near market-neutral after hedging
- **CMA β: 0.18 → 0.43** ($p<0.05$) — hedging amplifies conservative-investment exposure 2.4×
- **R^2: 0.91 → 0.17** — hedged alpha NOT explained by standard FF5 + Mom factors
- **Mechanism:** Quality + Low Vol selects mature, low-capex firms → naturally CMA-positive
- **Implication:** "Free" factor diversification — emergent CMA adds value beyond targeted factors

Source: FF5+Mom OOS regression, Holdout 2019–2024 | *** p<0.01, ** p<0.05 | EM FF factors

05

US Portfolio Diversification

Does EM alpha add value to an existing US allocation?

Alpha survives incremental factor controls; ρ = 0.22 with US factors

INCREMENTAL FACTOR REGRESSIONS: TO_MVO Hedged

	CAPM	FF3	FF5	FF5+Mom
const	0.0036	0.0034	0.0025	0.0025
	(0.0024)	(0.0025)	(0.0025)	(0.0027)
Mkt_RF	0.0816*	0.0870**	0.1040***	0.0998**
	(0.0448)	(0.0435)	(0.0391)	(0.0447)
SMB		−0.0210	0.0777	0.0668
		(0.0958)	(0.1049)	(0.1216)
HML		−0.0313	−0.1880**	−0.1926**
		(0.0596)	(0.0749)	(0.0781)
RMW			0.1089	0.1031
			(0.1052)	(0.1185)
CMA			0.2932**	0.3026***
			(0.1239)	(0.1138)
Mom				−0.0188
				(0.1135)
R-squared	0.0499	0.0584	0.1571	0.1581
R-squared Adj.	0.0357	0.0150	0.0902	0.0766
Adj R^2	0.036	0.015	0.090	0.077
N	69	69	69	69
R^2	0.050	0.058	0.157	0.158

Standard errors in parentheses.
* p<.1, ** p<.05, ***p<.01

Key Findings

- **α positive across all models:** CAPM (0.36%), FF3 (0.34%), FF5 (0.25%), FF5+Mom (0.25%) — survives incremental factor controls
- **CMA: 0.29→0.30** (p<0.01 in FF5+Mom) — only consistently significant factor loading besides Market
- **R^2: 0.05→0.16** — even FF5+Mom explains only 16% of hedged returns; most alpha is unexplained
- **US Factor ρ: 0.22** — near-zero correlation with SMB, HML, RMW, CMA, Mom; genuine diversifier for global portfolios
- **Implication:** EM factor premia are not redundant with US/global factors — strategy adds diversification when you need it most

Corr. with US Factors	Mkt-RF	SMB	HML	RMW	CMA	Mom	US Mkt
TO_MVO Hedged	0.22	0.01	−0.08	0.10	0.10	−0.05	0.22

Source: NB19 | Incremental EM FF regressions + US factor correlations | Holdout 2019–2024 | * p<0.1, ** p<0.05, *** p<0.01

Hurdle Sharpe premium of 0.64 makes EM a compelling portfolio addition

```
Expected Returns (annualized):
  US Market:     15.23%
  EM Hedged:     5.52%

Correlation: 0.223

Annualized Covariance Matrix:
           US_Mkt  EM_Hedged
US_Mkt     0.0332     0.0027
EM_Hedged  0.0027     0.0044

Portfolio Metrics at Various EM Allocations (lambda=2.5):
 EM Weight  US Weight    Ann Ret    Ann Vol   Sharpe   Vol Change
-----------------------------------------------------------------
        0%       100%     15.23%     18.23%   0.835       +0.0%
        5%        95%     14.74%     17.39%   0.847       -4.6%
       10%        90%     14.26%     16.57%   0.860       -9.1%
       15%        85%     13.77%     15.75%   0.874      -13.6%
       20%        80%     13.28%     14.94%   0.889      -18.1%
       25%        75%     12.80%     14.14%   0.905      -22.4%
       50%        50%     10.37%     10.38%   0.999      -43.1%
```

Hurdle Sharpe Ratio

$$SR_{hurdle} = SR_{US} \times \rho(\text{US, EM})$$
$$= 0.835 \times 0.223 = 0.186$$

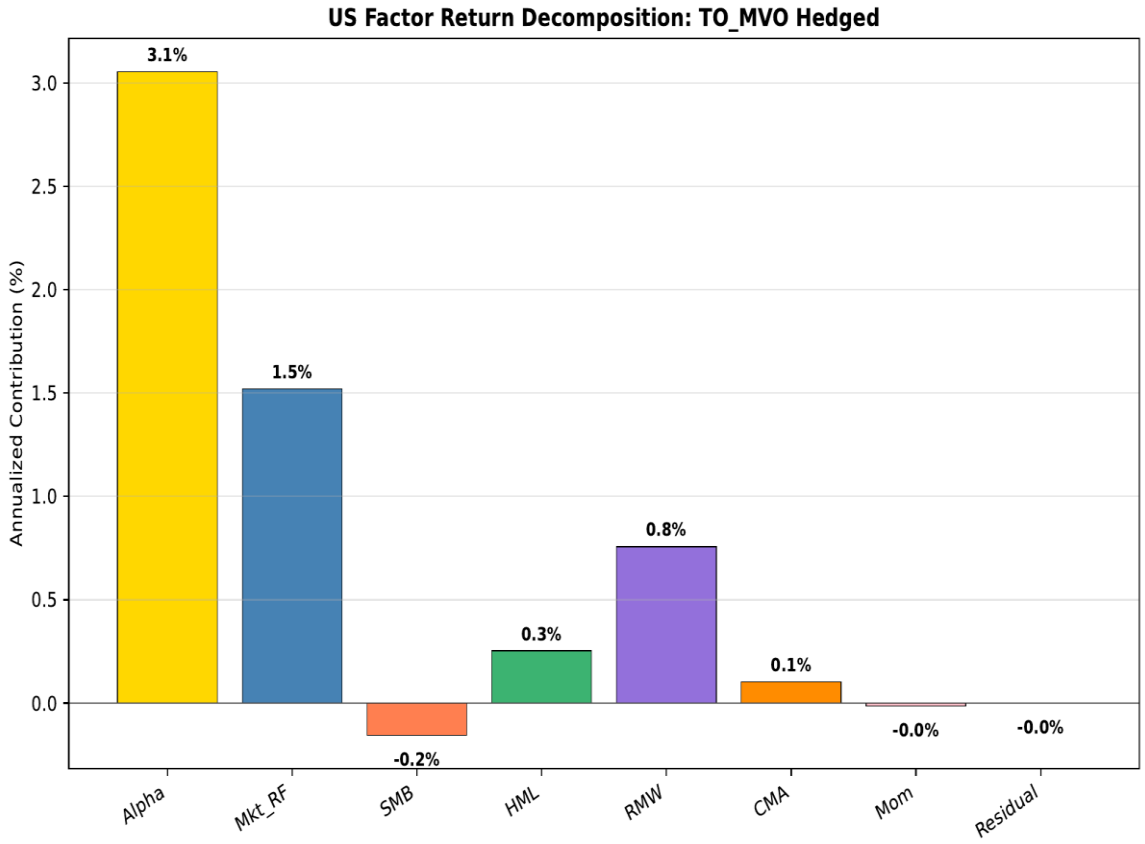
The minimum Sharpe ratio the new asset must deliver to improve the existing portfolio's risk-adjusted return. If the new asset's Sharpe exceeds the hurdle, adding it is beneficial regardless of its standalone return level.

Key Findings

- **SR_{hurdle}: 0.186** — low bar set by ρ = 0.223 with US market
- **$SR_{EM\ Hedged}$: 0.832** — clears the hurdle by 4.5× margin
- **Premium: 0.649** — driven by low correlation + hedged alpha
- **SR gain:** — 0.835 (0%) → 0.905 (25%) → 1.0 (50%)
- **TC adjusted SR: 0.672** — still beat the hurdle 0.186
- **Vol reduction scales:** — 18.2% → 10.4% (50% EM), −43% at half-weight

Source: Out-of-sample period 2019–2024. Annualized covariance from monthly returns. λ = 2.5 (mean-variance framework).

Alpha is 56% of hedged return; US factors explain only 16% of variance



Performance Decomposition

$R_{hedged} = \alpha + \Sigma\, \beta_k \times F^{US}_k + \varepsilon$

Regress hedged returns on US FF5+Mom factors. Each bar = β × mean factor return, annualized. Alpha is the unexplained intercept.

Key Findings

- **α: 3.06% ann. (306 bps) — 56% of total 5.52% return**
- **Mkt: 1.52%** — residual market beta contributes **28%** of return
- **RMW: 0.76%** — profitability factor is largest positive tilt (**14%**)
- **HML + CMA: 0.4%** — value and investment add modest **6%** of return
- **SMB: −0.16%** — slight large-cap tilt creates small drag
- **Mom: 0%** — almost no momentum exposure
- **R^2: 0.158** — US factors explain only 16% of hedged return variance

Source: OLS regression on US FF5+Mom (OOS 2019–2024, 69 months). HC1 robust standard errors.

Conclusions

1. Parsimonious model, real out-of-sample alpha

EW composite + TO_MVO → Sharpe 0.67, α = 4.5% net of 45 bps TC (holdout 2019–2024).

2. Dynamic hedge compresses risk, preserves alpha

Max DD −27% → −5%, vol −65%. Survives COVID and 2022 EM drawdown.

3. Orthogonal to US factors — 84% of variance is new information

ρ = 0.22, FF5+Mom R^2 = 0.16. Alpha is 56% of total return.

4. For US investors, EM factor alpha is a natural portfolio diversifier

Clears hurdle Sharpe by 4.5×. Even a small EM weight improves the portfolio's risk-adjusted frontier.

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